

OFFICIAL APPLICATION MANUAL & REFERENCE

NODEBOX

A Comprehensive Encyclopedia of Procedural Geometry,
Generative Art, Pattern Design, Audio Synthesis & Node Recipes

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GPU Accelerated

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1. Architecture & Working Principles

NodeBox is a modular, high-performance node-based environment for generative vector art, graphic design, procedural pattern creation, and data visualization. Unlike imperative scripting where sequences of commands run once top-to-bottom, NodeBox computes graphics using a **Directed Acyclic Graph (DAG)**.

Core Concepts

- **Nodes:** Independent, pure, reusable functional units. A node accepts input ports, applies mathematical transformations or vector algorithms, and delivers output data.
- **Ports & Data Types:** Ports are strongly typed channels. Standard types include geometry (vector paths & shapes), color (RGBA and color palettes), point (2D coordinates), float / int (numerical values), string (text data), and list (collections).
- **Automatic List Looping:** When a node with a single-value input receives a list of values, NodeBox automatically loops over the list and yields a list of results. For instance, connecting a list of 100 points into an ellipse node generates 100 ellipses without any explicit loop statements.
- **Rendered Node vs. Active Node:**
 - **Rendered Node** (Flagged with the solid circle): Dictates what is rendered to the canvas viewport.
 - **Active Node** (Highlighted with a blue border): Dictates what parameters are displayed in the Inspector Pane for interactive adjustment.

2. Canvas Viewport & Modal Controls

The canvas viewport combines real-time GPU-accelerated rendering with fast physical transformation tools:

- **Blender-Style Modal Transformations:**
 - Press **G** to enter **Grab / Move** mode. Move mouse to translate; click or press **Enter** to confirm; right-click or **Esc** to cancel.
 - Press **R** to enter **Rotate** mode. Move mouse radially to set angle.
 - Press **S** to enter **Scale** mode.
 - While in G, R, or S: press **X** or **Y** to constrain transformation to the horizontal or vertical axis. Hold **Shift** for fine precision snap.
- **Canvas Alt + Drag Quick Value Scrubbing:**
 - Hold **Alt** and drag horizontally with the left mouse button on the canvas to scrub the primary numerical parameter of the active node (e.g. count, radius, spacing, amplitude).
 - Hold **Shift** while dragging for decimal fine-tuning.
- **Infinite 3×3 Wallpaper Repeat Viewport Mode (**T**):**
 - Press **T** or toggle **Repeat 3×3** in the header bar. The viewport displays 8 ghosted adjacent tiles surrounding the center canvas, separated by an amber dashed boundary grid. This allows textile and surface designers to preview edge-wrapping seamless repeats in real time.

3. Complete Node Library Reference

This reference lists all 193 built-in and extended nodes across all NodeBox libraries, complete with parameter tables, default values, and operational guidelines.

color.color

Prototype: core.node

Output: color

Create a color value that can be used as a variable.

Port	Type	Default	Description
color	color	#000000ff	Sets the color parameter.

color.gray_color

Prototype: core.node

Output: color

Create a gray color.

Port	Type	Default	Description
gray	float	0.0	Sets the gray parameter.
alpha	float	255.0	Sets the alpha parameter.
range	float	255.0	Sets the range parameter.

color.hsb_color

Prototype: core.node

Output: color

Create a HSB color.

Port	Type	Default	Description
hue	float	0.0	Sets the hue parameter.
saturation	float	0.0	Sets the saturation parameter.
brightness	float	0.0	Sets the brightness parameter.
alpha	float	255.0	Sets the alpha parameter.
range	float	255.0	Sets the range parameter.

color.rgb_color

Prototype: core.node

Output: color

Create a RGB color.

Port	Type	Default	Description
red	float	0.0	Sets the red parameter.
green	float	0.0	Sets the green parameter.
blue	float	0.0	Sets the blue parameter.
alpha	float	255.0	Sets the alpha parameter.
range	float	255.0	Sets the range parameter.

color.palette

Prototype: core.node

Output: color

Generate a color harmony palette.

Port	Type	Default	Description
baseColor	color	#0077ffff	Sets the baseColor parameter.
type	string	analogous	Sets the type parameter.
count	int	5	Sets the count parameter.
spread	float	0.08	Sets the spread parameter.

color.gradient

Prototype: core.node

Output: color

Sample an interpolated color along a multi-stop gradient using Oklab perceptual color space.

Port	Type	Default	Description
colors	list	–	List of gradient stop colors.
position	float	0.5	Position along the gradient (0.0 - 1.0 or 0 - 100).
colorSpace	string	oklab	Interpolation color space.

color.extract_palette

Prototype: core.node

Output: color

Extract dominant color palette swatches from an image using K-Means clustering.

Port	Type	Default	Description
imageFile	string	—	Path to the image file.
count	int	5	Number of dominant colors to extract.
sampleStep	int	4	Pixel sampling resolution step.

color.blend_mode

Prototype: core.node

Output: color

Blend two colors using standard graphic design blend modes.

Port	Type	Default	Description
baseColor	color	#ffffffff	Base background color.
blendColor	color	#ff0077ff	Overlay blend color.
mode	string	multiply	Graphic design blend mode.
opacity	float	1.0	Blend opacity (0.0 - 1.0).

color.conic_gradient

Prototype: core.node

Output: color

Generate an angular / conical gradient sweep of colors around a center point.

Port	Type	Default	Description
colors	list	—	List of gradient stop colors.
steps	int	100	Number of color steps around the sweep.
startAngle	float	0.0	Sweep start angle in degrees.
endAngle	float	360.0	Sweep end angle in degrees.
space	string	oklab	Interpolation color space.

CORE Library

1 Nodes

core.frame

Prototype: core.node

Output: float

Output the value of the current frame. Used for animating ports.

Port	Type	Default	Description
context	context	—	The internal execution context from which to extract the frame number.

COREVECTOR Library

78 Nodes

corevector.generator

Prototype: core.node

Output: geometry

Template for creating nodes that generate vector data.

No input ports required (standalone generator).

corevector.filter

Prototype: generator

Output: generator

Template for creating nodes that manipulate vector data.

Port	Type	Default	Description
shape	geometry	—	The input shape.

corevector.align

Prototype: filter

Output: filter

Align a shape in relation to the origin.

Port	Type	Default	Description
position	point	0.00,0.00	The alignment point.
halign	string	center	The horizontal alignment.
valign	string	middle	The vertical alignment.

corevector.arc

Prototype: generator

Output: generator

Create an arc.

Port	Type	Default	Description
position	point	0.00,0.00	The center point of the arc.
width	float	100.0	The width of the arc.
height	float	100.0	The height of the arc.
start_angle	float	0.0	The starting angle of the arc. 0 degrees is at 3 o'clock.
degrees	float	45.0	The amount of degrees of the arc.
type	string	pie	The type of arc: pie, chord or open.

corevector.centroid

Prototype: filter

Output: point

Calculate the geometric center point of a shape.

No input ports required (standalone generator).

corevector.colorize

Prototype: filter

Output: filter

Change the color of a shape.

Port	Type	Default	Description
fill	color	#000000ff	The new fill color. Set alpha to 0 for no fill.
stroke	color	#000000ff	The new stroke color.
strokeWidth	float	0.0	The new stroke width. Set to 0 for no stroke.

corevector.compound

Prototype: generator

Output: generator

Add, subtract or intersect geometry.

Port	Type	Default	Description
shape1	geometry	—	The first shape.
shape2	geometry	—	The second shape.
function	string	united	The method to use for combining the shapes.
invert_difference	boolean	false	Switch shape1 and shape2 internally.

corevector.connect

Prototype: generator

Output: generator

Connect all points in a path.

Port	Type	Default	Description
points	list	–	The list of points to connect.
closed	boolean	true	If true, closes the shape. This has an effect on the path stroke.

corevector.copy

Prototype: filter

Output: filter

Create multiple copies of a shape.

Port	Type	Default	Description
copies	int	1	The amount of copies.
order	string	tsr	The order in which to perform translation, rotation and scale.
translate	point	0.00,0.00	The amount to move each copy.
rotate	float	0.0	The amount to rotate each copy.
scale	point	100.00,100.00	The amount to scale each copy.

corevector.delete

Prototype: filter

Output: filter

Delete points or paths that lie within the given bounding shape.

Port	Type	Default	Description
bounding	geometry	–	The bounding shape.
scope	string	points	Whether to delete points or paths.
operation	string	selected	Whether to delete the selected or non-selected points or paths.

corevector.distribute

Prototype: generator

Output: generator

Distribute shapes on a horizontal or vertical axis.

Port	Type	Default	Description
shapes	geometry	–	The list of shapes to distribute.
horizontal	string	none	The method of horizontal distribution.
vertical	string	none	The method of vertical distribution.

corevector.ellipse

Prototype: generator

Output: generator

Create an ellipse or circle.

Port	Type	Default	Description
position	point	0.00,0.00	The center point of the ellipse.
width	float	100.0	The width of the ellipse.
height	float	100.0	The height of the ellipse.

corevector.fit

Prototype: filter

Output: filter

Fit a shape within bounds.

Port	Type	Default	Description
position	point	0.00,0.00	The target center point of the shape.
width	float	300.0	The maximum target width of the shape.
height	float	300.0	The maximum target height of the shape.
keep_proportions	boolean	true	If true, does not squash or stretch the shape.

corevector.fit_to

Prototype: filter

Output: filter

Fit a shape to another shape.

Port	Type	Default	Description
bounding	geometry	–	The bounding shape to take position, width and height from.
keep_proportions	boolean	true	If true, does not squash or stretch the shape.

corevector.freehand

Prototype: generator

Output: generator

Draw directly on the canvas using the mouse.

Port	Type	Default	Description
path	string	—	The internal path data.

corevector.grid

Prototype: generator

Output: point

Create a grid of points.

Port	Type	Default	Description
columns	int	10	The amount of columns in the grid.
rows	int	10	The amount of rows in the grid.
width	float	300.0	The total width of the grid.
height	float	300.0	The total height of the grid.
position	point	0.00,0.00	The center point of the grid.

corevector.group

Prototype: generator

Output: generator

Combine multiple geometries together.

Port	Type	Default	Description
shapes	geometry	—	The list of shapes to group.

corevector.import_svg

Prototype: generator

Output: generator

Import geometry from a SVG file.

Port	Type	Default	Description
file	string	—	The SVG file to import.
centered	boolean	true	If true, centers the shape around the position.
position	point	0.00,0.00	The target position of the shape.

corevector.line

Prototype: generator

Output: generator

Create a line between two points.

Port	Type	Default	Description
point1	point	0.00,0.00	The starting point of the line.
point2	point	100.00,100.00	The ending point of the line.
points	int	2	The amount of points to generate along the line.

corevector.line_angle

Prototype: generator

Output: generator

Create a line between one point and an angle + distance.

Port	Type	Default	Description
position	point	0.00,0.00	The starting point of the line.
angle	float	0.0	The direction of the line.
distance	float	100.0	The distance between starting and ending point.
points	int	2	The amount of points to generate along the line.

corevector.link

Prototype: generator

Output: generator

Generate a visual link between two shapes.

Port	Type	Default	Description
shape1	geometry	—	The first bounding volume of the link shape.
shape2	geometry	—	The second bounding volume of the link.
orientation	string	horizontal	The direction of the link shape.

corevector.make_point

Prototype: generator

Output: point

Create a point from X/Y coordinates.

Port	Type	Default	Description
x	float	0.0	The X position of the point.
y	float	0.0	The Y position of the point.

corevector.null

Prototype: filter

Output: filter

Do nothing.

No input ports required (standalone generator).

corevector.point

Prototype: core.node

Output: point

Create a point value that can be used as a variable.

Port	Type	Default	Description
value	point	0.00,0.00	The input point.

corevector.point_on_path

Prototype: generator

Output: generator

Calculate a point on a path.

Port	Type	Default	Description
shape	geometry	—	The input shape.
t	float	0.0	The relative position on the shape (0 - 100).

corevector.polygon

Prototype: generator

Output: generator

Create a multi-sided polygon.

Port	Type	Default	Description
position	point	0.00,0.00	The center point of the polygon.
radius	float	100.0	The size of the polygon.
sides	int	3	The number of sides the polygon has.
align	boolean	false	If true, aligns the polygon to the X axis.

corevector.quad_curve

Prototype: generator

Output: generator

Create a quadratic curve with one off-curve point.

Port	Type	Default	Description
point1	point	0.00,0.00	The starting point.
point2	point	100.00,0.00	The ending point.
t	float	50.0	The position of the off-curve point (0 - 100).
distance	float	50.0	The offset of the curve from a straight line.

corevector.rect

Prototype: generator

Output: generator

Create a rectangle, square or rounded rectangle.

Port	Type	Default	Description
position	point	0.00,0.00	The center point of the rectangle.
width	float	100.0	The width of the rectangle.
height	float	100.0	The height of the rectangle.
roundness	point	0.00,0.00	The roundness of the rectangle.

corevector.reflect

Prototype: filter

Output: filter

Mirror the geometry around an invisible axis.

Port	Type	Default	Description
position	point	0.00,0.00	The center point of the mirror operation.
angle	float	120.0	The angle at which to mirror the shape.
keep_original	boolean	true	If true, also retains the original geometry.

corevector.resample

Prototype: filter

Output: filter

Distribute points along a shape.

Port	Type	Default	Description
method	string	length	The method of distribution.
length	float	10.0	The maximum length of each segment.
points	int	10	The amount of output points.
per_contour	boolean	false	If true, the given amount is per contour, not for the whole path.

corevector.rotate

Prototype: filter

Output: filter

Rotate the shape according to the given angle.

Port	Type	Default	Description
angle	float	0.0	The rotation angle (in degrees).
origin	point	0.0,0.0	The point around which to rotate.

corevector.round_segments

Prototype: filter

Output: filter

Transform the straight corners of a shape into curvy ones.

Port	Type	Default	Description
d	float	10.0	The distance from the control points to the corner point.

corevector.scale

Prototype: filter

Output: filter

Resize the shape by scaling it.

Port	Type	Default	Description
scale	point	100.00,100.00	The scale factor.
origin	point	0.0,0.0	The point around which to scale.

corevector.scatter

Prototype: filter

Output: point

Generate points within the boundaries of a shape.

Port	Type	Default	Description
amount	int	20	The amount of output points to generate.
seed	int	0	The random variation.

corevector.shape_on_path

Prototype: generator

Output: generator

Copies shapes on a path.

Port	Type	Default	Description
shape	geometry	–	The shape(s) to copy.
path	geometry	–	The path on which to copy the shape(s).
amount	int	1	The amount of copies.
alignment	string	leading	The way to lay out the shapes horizontally.
spacing	float	20.0	The spacing between the shapes.
margin	float	0.0	Empty space before and/or after the shapes.
baseline_offset	float	0.0	The vertical offset of the shapes from the baseline.

corevector.skew

Prototype: filter

Output: filter

Skew the shape.

Port	Type	Default	Description
skew	point	0.0,0.0	The skew factor.
origin	point	0.0,0.0	The point around which to skew.

corevector.snap

Prototype: filter

Output: filter

Snap geometry to a grid.

Port	Type	Default	Description
distance	float	10.0	The size of the grid cells.
strength	float	100.0	The strength of attraction to the grid (0=no influence, 100=all points are strictly on the grid).
position	point	0.00,0.00	The position of the grid.

corevector.sort

Prototype: core.node

Output: custom

Sort points or shapes using different sorting methods.

Port	Type	Default	Description
shapes	geometry	–	The list of shapes to sort.
order_by	string	none	The sort method.
position	point	0.00,0.00	The center point.

corevector.stack

Prototype: core.node

Output: geometry

Arrange shapes in a horizontal or vertical layout.

Port	Type	Default	Description
shapes	geometry	–	Sets the shapes parameter.
direction	string	e	Sets the direction parameter.
margin	float	5.0	Sets the margin parameter.

corevector.star

Prototype: generator

Output: generator

Create a star shape.

Port	Type	Default	Description
position	point	0.00,0.00	The center point of the star.
points	int	20	The amount of points in the star.
outer	float	200.0	The outer diameter of the star.
inner	float	100.0	The inner diameter of the star.

corevector.text_on_path

Prototype: generator

Output: generator

Create a text path that follows a shape or curve.

Port	Type	Default	Description
text	string	text following a path	The input text.
path	geometry	—	The path on which to put the text.
font_name	string	Verdana	The name of the font.
font_size	float	24.0	The size of the font.
alignment	string	LEFT	Text alignment along the path.
margin	float	0.0	The relative starting position of the text on the path (0 - 100).
baseline_offset	float	0.0	The vertical offset of the text from the path.
tracking	float	0.0	Letter tracking/spacing offset.
flip	boolean	false	Flip text to the opposite/inside of the curve.

corevector.textpath

Prototype: generator

Output: generator

Create a path out of text.

Port	Type	Default	Description
text	string	hello	The input text.
font_name	string	Verdana	The name of the font.
font_size	float	24.0	The size of the font.
align	string	CENTER	The alignment of the text.
position	point	0.00,0.00	The center point of the text.
width	float	0.0	The maximum width of the text.

corevector.translate

Prototype: filter

Output: filter

Move the shape, changing its position.

Port	Type	Default	Description
translate	point	0.00,0.00	The amount of translation.

corevector.ungroup

Prototype: filter

Output: filter

Decompose the input geometry into its paths.

No input ports required (standalone generator).

corevector.wiggle

Prototype: filter

Output: filter

Shift points by a random amount.

Port	Type	Default	Description
scope	string	points	The type of objects to wiggle: points, contour or paths.
offset	point	10.00,10.00	The maximum amount of translation.
seed	int	0	The random variation.

corevector.delaunay

Prototype: generator

Output: geometry

Compute a Delaunay triangulation from a set of points.

Port	Type	Default	Description
points	list	–	The input points to triangulate.

corevector.voronoi

Prototype: generator

Output: geometry

Generate a Voronoi diagram from points with optional cell inset.

Port	Type	Default	Description
points	list	–	The seed points for the Voronoi cells.
width	float	500.0	The bounding width of the diagram.
height	float	500.0	The bounding height of the diagram.
inset	float	0.0	Centroid inset factor (0 = no inset, 0.2 = margin between cells).

corevector.convex_hull

Prototype: generator

Output: geometry

Compute the convex hull bounding polygon around a set of points.

Port	Type	Default	Description
points	list	–	The input points.

corevector.smooth

Prototype: filter

Output: filter

Smooth polygonal curves using Chaikin's corner-cutting algorithm.

Port	Type	Default	Description
iterations	int	3	Number of Chaikin subdivision iterations.
ratio	float	0.25	Corner cut ratio (0.25 is classic Chaikin).

corevector.flow_field

Prototype: generator

Output: geometry

Generate organic vector streamlines guided by a Simplex noise vector field.

Port	Type	Default	Description
seedPoints	list	–	Seed points from which streamlines originate (e.g. grid or scatter).
steps	int	50	Number of step segments per streamline.
stepLength	float	5.0	Length of each step segment.
noiseScale	float	0.005	Spatial frequency / scale of the noise field.
angleStrength	float	6.283185307179586	Angular multiplier (2*PI maps noise [-1, 1] to a full circle).
seed	int	0	Random seed variation.

corevector.lsystem

Prototype: generator

Output: geometry

Generate fractal branching structures and curves using an L-System.

Port	Type	Default	Description
axiom	string	F	The initial axiom string.
rules	string	F=F+F-F-F+F	Grammar rewrite rules (e.g. F=F+F-F-F+F or multi-rule separated by semicolon/newline).
generations	int	3	The number of recursive rewriting generations.
angle	float	90.0	The turning angle in degrees for + and -.
length	float	10.0	The base step length.
length_scale	float	1.0	Multiplier applied to length per generation.
position	point	0.00,0.00	Starting root position.

corevector.attractor

Prototype: filter

Output: filter

Attract, repel, or swirl points and vector geometry around a center point.

Port	Type	Default	Description
position	point	0.00,0.00	The center attractor / repulsor position.
force	float	50.0	Force magnitude. Positive attracts, negative repels.
radius	float	200.0	Maximum influence radius.
mode	string	linear	Force attenuation behavior.

corevector.offset_path

Prototype: filter

Output: filter

Expand or shrink a vector path by an offset distance.

Port	Type	Default	Description
distance	float	10.0	Offset distance. Positive expands, negative shrinks.
join	string	round	Corner join style.

corevector.trace_contours

Prototype: generator

Output: geometry

Extract vector isoline contours from an image at a brightness threshold.

Port	Type	Default	Description
file	string	—	Image file to trace.
threshold	float	50.0	Brightness threshold isoline level (0 - 100).
resolution	int	50	Sampling grid resolution.
position	point	0.00,0.00	Center position of traced contours.
width	float	500.0	Output width.
height	float	500.0	Output height.

corevector.typeset

Prototype: generator

Output: geometry

Format and wrap text into rectangular editorial multi-column layouts.

Port	Type	Default	Description
text	string	Typeset text layout	The input text to typeset.
fontName	string	Helvetica	The font family name.
fontSize	float	14.0	Font size in points.
alignment	string	LEFT	Text alignment.
position	point	0.00,0.00	Top-left origin of the text box.
width	float	400.0	Total bounding width.
height	float	0.0	Total height (0 = unbounded).
lineHeight	float	1.4	Line height multiplier (leading).
tracking	float	0.0	Letter spacing adjustment.
columns	int	1	Number of text columns.
columnGutter	float	20.0	Gutter spacing between columns.

corevector.fit_text

Prototype: generator

Output: geometry

Automatically calculate font size to fit text snugly into a bounding box.

Port	Type	Default	Description
text	string	HEADLINE	The input text.
fontName	string	Helvetica	The font name.
position	point	0.00,0.00	Center position of text.
width	float	300.0	Maximum allowable width.
height	float	80.0	Maximum allowable height.
minFontSize	float	8.0	Minimum font size floor.
maxFontSize	float	200.0	Maximum font size ceiling.
alignment	string	CENTER	Text alignment.

corevector.halftone

Prototype: generator

Output: geometry

Generate scalable vector halftone dot, line, or cross-hatch screens from an image.

Port	Type	Default	Description
imageFile	string	–	Path to raster image file.
position	point	0.00,0.00	Center position of halftone screen.
width	float	0.0	Screen width (0 = image width).
height	float	0.0	Screen height (0 = image height).
dotSpacing	float	12.0	Screen grid spacing between dots/lines.
angle	float	45.0	Screen rotation angle in degrees.
shapeType	string	dot	Halftone screen element shape.
scale	float	1.0	Dot/line size multiplier.
threshold	float	0.0	Minimum luminance cutoff threshold.
invert	boolean	false	Invert positive/negative screen.

corevector.truchet_tiles

Prototype: generator

Output: geometry

Generate procedural seamless decorative tiling patterns (arcs, diagonals, triangles).

Port	Type	Default	Description
position	point	0.00,0.00	Center position of the tiling.
width	float	400.0	Total pattern width.
height	float	400.0	Total pattern height.
tileSize	float	40.0	Size of each individual tile unit.
tileType	string	arcs	Tiling algorithm pattern.
lineWidth	float	2.0	Stroke width for line/arc patterns.
seed	int	1	Random permutation seed.

corevector.guilloche

Prototype: generator

Output: geometry

Generate ornate mathematical spirograph guilloche rosettes and security borders.

Port	Type	Default	Description
position	point	0.00,0.00	Center position.
majorRadius	float	120.0	Radius of fixed circle.
minorRadius	float	40.0	Radius of rolling circle.
offset	float	50.0	Drawing pen distance from center of rolling circle.
steps	float	2000.0	Number of curve interpolation steps.
revolutions	float	10.0	Number of complete circular revolutions.
modulationAmp	float	0.0	Harmonic amplitude modulation.
modulationFreq	float	0.0	Harmonic frequency modulation.

corevector.metaballs

Prototype: generator

Output: geometry

Blend neighboring circles into smooth liquid gooey metaball shapes.

Port	Type	Default	Description
circles	list	–	List of circles (paths or points).
threshold	float	30.0	Maximum bridging distance gap between circles.
handleFactor	float	0.5	Smoothness of connecting tangential Bezier bridges.
unite	boolean	true	Unite circles and bridges into a single solid path.

corevector.modular_grid

Prototype: generator

Output: geometry

Generate a Swiss/Bauhaus editorial modular grid with columns, rows, gutters, and margins.

Port	Type	Default	Description
position	point	0.00,0.00	Center position of grid.
width	float	800.0	Total layout width.
height	float	600.0	Total layout height.
columns	int	4	Number of layout columns.
rows	int	3	Number of layout rows.
colGutter	float	15.0	Gutter spacing between columns.
rowGutter	float	15.0	Gutter spacing between rows.
topMargin	float	40.0	Outer top margin.
bottomMargin	float	40.0	Outer bottom margin.
leftMargin	float	40.0	Outer left margin.
rightMargin	float	40.0	Outer right margin.
outputMode	string	cells	Grid output mode.

corevector.align_distribute

Prototype: generator

Output: geometry

Align and distribute multiple shapes with uniform edge-to-edge spacing.

Port	Type	Default	Description
shapes	list	–	Shapes to align and distribute.
align	string	none	Alignment mode.
distribute	string	none	Distribution mode.
spacing	float	10.0	Gap between shapes when distributing.

corevector.pack_circles

Prototype: generator

Output: geometry

Tightly pack circles of varying radii into a container without overlap.

Port	Type	Default	Description
radii	list	—	List of circle radii.
center	point	0.00,0.00	Center anchor point.
containerRadius	float	200.0	Optional outer container radius (0 = free packing).
iterations	int	150	Relaxation simulation iterations.
seed	int	1	Initial distribution seed.

corevector.roughen

Prototype: filter

Output: filter

Add organic hand-drawn wobble, pencil jitter, or deckled edges to vector paths.

Port	Type	Default	Description
size	float	5.0	Displacement magnitude in points.
detail	float	2.0	Subdivision detail frequency along path.
method	string	smooth	Jitter noise algorithm.
seed	int	1	Random variation seed.

corevector.long_shadow

Prototype: filter

Output: filter

Project flat geometric long shadows from vector shapes at an angle.

Port	Type	Default	Description
angle	float	45.0	Direction angle in degrees.
distance	float	50.0	Shadow projection length.
fade	float	0.0	Shadow transparency falloff (0 - 1).

corevector.extrude_3d

Prototype: filter

Output: filter

Extrude 2D vector shapes into pseudo-3D isometric facets with realistic face shading.

Port	Type	Default	Description
depth	float	20.0	Axonometric extrusion depth.
angle	float	45.0	Extrusion angle in degrees.
frontColor	color	#dcdce6ff	Front cap color.
sideColor	color	#595966ff	Base color for extruded side facets.
lightIntensity	float	80.0	Directional lighting contrast.

corevector.boolean

Prototype: generator

Output: generator

Perform 2D Constructive Solid Geometry (CSG) boolean operations between two shapes.

Port	Type	Default	Description
shape1	geometry	–	Primary shape.
shape2	geometry	–	Secondary shape.
operation	string	union	Boolean CSG operation.

corevector.stroke_style

Prototype: filter

Output: filter

Customize stroke styles with dashes, dash phase, caps, joins, and arrowheads.

Port	Type	Default	Description
dashLength	float	0.0	Length of dash segments (0 = solid).
gapLength	float	0.0	Length of gaps between dashes.
dashPhase	float	0.0	Dash animation phase offset.
cap	string	butt	End cap style.
join	string	miter	Corner join style.
startArrow	boolean	false	Add arrowhead at start of path.
endArrow	boolean	false	Add arrowhead at end of path.
arrowSize	float	10.0	Size of arrowheads in points.

corevector.morph

Prototype: generator

Output: geometry

Smoothly morph and tween between two arbitrary vector shapes.

Port	Type	Default	Description
shapeA	geometry	–	Initial shape (progress = 0.0).
shapeB	geometry	–	Target shape (progress = 1.0).
progress	float	0.5	Interpolation progress slider (0.0 to 1.0).
samples	int	100	Number of perimeter sample vertices.

corevector.artboard

Prototype: generator

Output: geometry

Define an export frame artboard for multi-canvas page layout and automated batch export.

Port	Type	Default	Description
shape	geometry	–	Vector shape content to place inside the artboard.
name	string	Artboard 1	Name of the artboard (used as export filename).
preset	string	custom	Predefined export dimension standard.
width	float	1080.0	Width in pixels/points.
height	float	1080.0	Height in pixels/points.
position	point	0.00,0.00	Artboard center position.
clip	boolean	true	Clip shapes extending beyond artboard boundaries.
background	color	#ffffff	Artboard backdrop fill color.
showFrame	boolean	true	Display border and name title in viewport.

corevector.round_corners

Prototype: generator

Output: geometry

Round or style sharp corners of shapes, polygons, and text outlines.

Port	Type	Default	Description
shape	geometry	–	Vector shape or text geometry.
radius	float	10.0	Corner radius in pixels.
type	string	round	Corner profile geometry.
threshold	float	175.0	Maximum interior corner angle to round.
clamp	boolean	true	Prevent corners from overlapping adjacent edges.

corevector.duplicator

Prototype: generator

Output: geometry

Cavalry-style procedural duplicator and cloner with step transforms and jitter.

Port	Type	Default	Description
shape	geometry	—	Shape to duplicate.
mode	string	linear	Clone distribution pattern.
count	int	5	Number of copies (linear, radial, spiral).
spacing	float	50.0	Linear clone distance.
angle	float	0.0	Linear clone direction in degrees.
columns	int	3	Grid column count.
rows	int	3	Grid row count.
spacingX	float	60.0	Grid horizontal spacing.
spacingY	float	60.0	Grid vertical spacing.
stagger	float	0.0	Horizontal offset for alternate rows (0.5 for brick pattern).
radius	float	120.0	Radial circle radius.
startAngle	float	0.0	Radial starting angle.
endAngle	float	360.0	Radial ending angle.
orient	boolean	true	Rotate clones to align with trajectory tangent.
growthRate	float	3.0	Spiral outward expansion rate.
angleStep	float	137.5	Spiral angle step (137.5 for golden ratio / phyllotaxis).
stepRotation	float	0.0	Cumulative rotation per clone in degrees.
stepScale	float	0.0	Cumulative scaling percentage per clone.
stepOpacity	float	1.0	Fade opacity across clones (1.0 = no fade).
jitterPos	float	0.0	Random displacement magnitude.
jitterRot	float	0.0	Random rotation variation.
jitterScale	float	0.0	Random scale variation percentage.
seed	int	1	Random variation seed.

corevector.seamless_tile

Prototype: generator

Output: geometry

Automatically wrap and duplicate vector shapes across tile edges to create seamless repeating patterns.

Port	Type	Default	Description
shape	geometry	—	Input vector shapes to tile.
width	float	400.0	Tile boundary width.
height	float	400.0	Tile boundary height.
clip	boolean	false	Clip outer edges strictly to tile boundary.
center	point	0.00,0.00	Center of the tile.

corevector.hex_grid

Prototype: generator

Output: geometry

Generate an interlocking hexagonal honeycomb lattice of polygon cells.

Port	Type	Default	Description
columns	int	5	Number of columns.
rows	int	5	Number of rows.
radius	float	50.0	Hexagon cell outer radius.
spacing	float	0.0	Gap spacing between hexagon cells.
orientation	string	pointy	Hexagon orientation.
center	point	0.00,0.00	Center of the hexagonal grid.

corevector.kaleidoscope

Prototype: generator

Output: geometry

Create N-fold rotational and bilateral mirror symmetry for mandalas, rosettes, and Moroccan tiles.

Port	Type	Default	Description
shape	geometry	—	Input shape motif to mirror.
segments	int	8	Number of symmetrical radial segments.
center	point	0.00,0.00	Center axis of symmetry.
mirror	boolean	true	Reflect alternate segments for bilateral symmetry.

corevector.wave_warp

Prototype: generator

Output: geometry

Deform vector shapes and typography along mathematical wave functions.

Port	Type	Default	Description
shape	geometry	—	Vector shape to warp.
axis	string	horizontal	Wave displacement axis.
waveType	string	sine	Mathematical wave profile.
amplitude	float	20.0	Displacement peak magnitude.
frequency	float	0.02	Wave cycle frequency.
phase	float	0.0	Wave phase offset in degrees.

corevector.bulge_pinch

Prototype: generator

Output: geometry

Radial lens deformer creating spherical bulge magnification or pinch distortion.

Port	Type	Default	Description
shape	geometry	—	Shape to deform.
center	point	0.00,0.00	Focal center of distortion.
radius	float	200.0	Radius of lens influence.
strength	float	0.6	Distortion magnitude (> 0 bulge, < 0 pinch).

corevector.stripes

Prototype: generator

Output: geometry

Generate parametric linear stripes, checkerboards, or sunburst rays.

Port	Type	Default	Description
width	float	600.0	Total pattern width.
height	float	600.0	Total pattern height.
count	int	10	Number of stripe divisions.
ratio	float	0.5	Stripe width vs gap balance.
angle	float	0.0	Pattern rotation angle in degrees.
type	string	linear	Stripe pattern layout.

data.import_text

Prototype: core.node

Output: string

Import a text file and return each line of the file.

Port	Type	Default	Description
file	string	–	The text file.

data.import_csv

Prototype: core.node

Output: data

Import a comma-separated value file containing tabular data.

Port	Type	Default	Description
file	string	–	The CSV file.
delimiter	string	comma	The character used to separate columns.
quotes	string	double	The type of quotes used in the file.
number_separator	string	period	The character used to separate the fractional part.

data.make_table

Prototype: core.node

Output: data

Make a data table out of lists.

Port	Type	Default	Description
headers	string	alpha;beta	The names of the headers separated by semicolons or commas.
list1	list	–	The first list.
list2	list	–	The second list.
list3	list	–	The third list.
list4	list	–	The fourth list.
list5	list	–	The fifth list.
list6	list	–	The sixth list.

data.lookup

Prototype: core.node

Output: data

Look up a value in a table or object.

Port	Type	Default	Description
list	list	–	The input value.
key	string	x	The key to lookup.

data.filter_data

Prototype: core.node

Output: data

Filter the data based on key/value.

Port	Type	Default	Description
data	data	–	Sets the data parameter.
key	string	name	Sets the key parameter.
op	string	=	Sets the op parameter.
value	string	Bob	Sets the value parameter.

data.sample_image

Prototype: core.node

Output: data

Sample pixel colors or brightness values from an image at given point coordinates.

Port	Type	Default	Description
file	string	–	The image file (PNG, JPG, GIF, BMP).
points	list	–	Points where the image should be sampled.
channel	string	brightness	The channel or value to sample.
origin	string	center	Coordinate origin mapping.

data.barchart

Prototype: core.node

Output: geometry

Generate a parametric vector bar chart from numbers.

Port	Type	Default	Description
values	list	–	Numeric data values.
position	point	0.00,0.00	Chart center position.
width	float	400.0	Total chart width.
height	float	200.0	Total chart height.
bar_padding	float	4.0	Padding between bars.
orientation	string	vertical	Bar orientation.

data.donut

Prototype: core.node

Output: geometry

Generate a parametric vector donut / pie chart from values.

Port	Type	Default	Description
values	list	–	Slice proportion values or percentages.
position	point	0.00,0.00	Donut center position.
inner_radius	float	50.0	Inner hole radius (0 for solid pie).
outer_radius	float	120.0	Outer rim radius.
start_angle	float	0.0	Starting angle in degrees.

DEVICE Library

10 Nodes

device.mouse_position

Prototype: core.node

Output: point

Get the most recent mouse position

Port	Type	Default	Description
context	context	–	Sets the context parameter.

device.receive_osc

Prototype: core.node

Output: string

Receive messages from OSC

Port	Type	Default	Description
device_name	string	osc1	The osc device to listen to.
address_prefix	string	/	The prefix of the addresses to receive the messages on.
arguments	string	–	The names of the OSC messages' arguments, delimited by ''
context	context	–	The internal execution context.

device.send_osc

Prototype: core.node

Output: custom

Send messages over OSC

Port	Type	Default	Description
address	string	–	Sets the address parameter.
port	int	9999	Sets the port parameter.
route	string	–	Sets the route parameter.
arguments	list	–	Sets the arguments parameter.

device.audio_analysis

Prototype: core.node

Output: float

Audio analysis with Minim

Port	Type	Default	Description
device_name	string	audio1	The audio device.
channel	string	mix	The channel to return values from.
averages	int	0	The number of averages to take.
context	context	–	The internal execution context.

device.audio_log_avg

Prototype: core.node

Output: float

Logarithmic FFT analysis with Minim

Port	Type	Default	Description
device_name	string	audio1	The audio device.
channel	string	mix	The channel to return values from.
baseFreq	int	22	The minimum frequency to include.
bandsPerOctave	int	3	The number of bands per octave.
context	context	–	The internal execution context.

device.audio_wave

Prototype: core.node

Output: float

Audio wave with Minim

Port	Type	Default	Description
device_name	string	audio1	The audio device.
context	context	–	The internal execution context.

device.beat_detect

Prototype: core.node

Output: data

Beat Detect with Minim

Port	Type	Default	Description
device_name	string	audio1	The audio device.
context	context	–	The internal execution context.

device.sound_file

Prototype: core.node

Output: float

Load sound files (mp3, wav, aif, flac, ogg) and extract normalized waveform samples.

Port	Type	Default	Description
file	string	–	Path to the sound file (mp3, wav, aif, etc.).
samples	int	100	Number of amplitude points to extract.
channel	string	mix	Audio channel.

device.sound_spectrum

Prototype: core.node

Output: float

Extract audio frequency spectrum band magnitudes from a sound file.

Port	Type	Default	Description
file	string	–	Path to the sound file (mp3, wav, aif, etc.).
bands	int	16	Number of frequency bands.

device.sound_info

Prototype: core.node

Output: data

Get metadata and volume metrics (duration, sample rate, RMS) for a sound file.

Port	Type	Default	Description
file	string	–	Path to the sound file.

LIST Library

19 Nodes

list.combine

Prototype: core.node

Output: list

Combine multiple lists into one.

Port	Type	Default	Description
list1	list	–	The first list.
list2	list	–	The second list.
list3	list	–	The third list.
list4	list	–	The fourth list.
list5	list	–	The fifth list.
list6	list	–	The sixth list.
list7	list	–	The seventh list.

list.count

Prototype: core.node

Output: int

Count the number of items in the list.

Port	Type	Default	Description
list	list	–	The input list.

list.cull

Prototype: core.node

Output: list

Remove all items from the list where the corresponding boolean is false.

Port	Type	Default	Description
list	list	–	The input list.
booleans	list	–	A list of matching boolean values. All list items where the booleans are true will be retained.

list.distinct

Prototype: core.node

Output: list

Remove all duplicate items from the list.

Port	Type	Default	Description
list	list	–	The input list.
key	string	–	An optional lookup key that will be searched for distinct values.

list.first

Prototype: core.node

Output: data

Take the first item of the list.

Port	Type	Default	Description
list	list	–	The input list.

list.keys

Prototype: core.node

Output: data

Take the keys of a map.

Port	Type	Default	Description
maps	list	–	The input map.

list.last

Prototype: core.node

Output: data

Take the last item of the list.

Port	Type	Default	Description
list	list	–	The input list.

list.pick

Prototype: core.node

Output: list

Pick items from the list in random order.

Port	Type	Default	Description
list	list	–	The input list.
amount	int	5	The amount of items to pick.
seed	int	0	The random variation.

list.repeat

Prototype: core.node

Output: list

Repeat the list a number of times.

Port	Type	Default	Description
list	list	–	The input list.
amount	int	1	The amount of times to repeat the list.
per_item	boolean	false	Repeat items one after another.

list.rest

Prototype: core.node

Output: list

Take all but the first item of the list.

Port	Type	Default	Description
list	list	–	The input list.

list.reverse

Prototype: core.node

Output: list

Reverse the list.

Port	Type	Default	Description
list	list	–	The input list.

list.second

Prototype: core.node

Output: data

Take the second item of the list.

Port	Type	Default	Description
list	list	–	The input list.

list.shift

Prototype: core.node

Output: list

Move items at the beginning of the list to the end.

Port	Type	Default	Description
list	list	–	The input list.
amount	int	1	The number of items to shift.

list.shuffle

Prototype: core.node

Output: list

Randomize the ordering of items in the list.

Port	Type	Default	Description
list	list	–	The input list.
seed	int	0	The random variation.

list.slice

Prototype: core.node

Output: list

Take a portion of the list.

Port	Type	Default	Description
list	list	–	The input list.
start_index	int	0	The starting index in the list.
size	int	10	The amount of items to take from the list.
invert	boolean	–	Omit the items instead of retaining them.

list.sort

Prototype: core.node

Output: list

Sort items in the list.

Port	Type	Default	Description
list	list	–	The input list.
key	string	–	The key used to compare and sort the items.

list.switch

Prototype: core.node

Output: custom

Switch between multiple inputs.

Port	Type	Default	Description
input1	list	–	The first list.
input2	list	–	The second list.
input3	list	–	The third list.
input4	list	–	The fourth list.
input5	list	–	The fifth list.
input6	list	–	The sixth list.
index	int	0	The index of the input list to select. If the index is greater than the number of inputs, it wraps around.

list.take_every

Prototype: core.node

Output: list

Take every nth element of the list.

Port	Type	Default	Description
list	list	–	The input list.
n	int	1	The amount of items to skip.

list.zip_map

Prototype: core.node

Output: data

Combine a list of keys and values together into a map.

Port	Type	Default	Description
keys	string	–	A list of keys.
values	list	–	A list of values.

math.abs

Prototype: core.node

Output: custom

Convert every number to a positive number.

Port	Type	Default	Description
value	float	0.0	Sets the value parameter.

math.add

Prototype: core.node

Output: custom

Add two numbers.

Port	Type	Default	Description
value1	float	0.0	Sets the value1 parameter.
value2	float	0.0	Sets the value2 parameter.

math.angle

Prototype: core.node

Output: custom

Calculate the angle between two points.

Port	Type	Default	Description
point1	point	0.00,0.00	Sets the point1 parameter.
point2	point	100.00,100.00	Sets the point2 parameter.

math.average

Prototype: core.node

Output: custom

Calculate the average of a list of numbers.

Port	Type	Default	Description
values	float	0.0	Sets the values parameter.

math.boolean

Prototype: core.node

Output: boolean

Create a boolean value that can be used as a variable.

Port	Type	Default	Description
value	boolean	false	Sets the value parameter.

math.ceil

Prototype: core.node

Output: custom

Round up the value of a number.

Port	Type	Default	Description
value	float	0.0	Sets the value parameter.

math.compare

Prototype: core.node

Output: custom

Select between two values by comparing them.

Port	Type	Default	Description
value1	float	0.0	Sets the value1 parameter.
value2	float	0.0	Sets the value2 parameter.
comparator	string	<	Sets the comparator parameter.

math.logical

Prototype: core.node

Output: boolean

Create a logical or operator.

Port	Type	Default	Description
boolean1	boolean	false	Sets the boolean1 parameter.
boolean2	boolean	false	Sets the boolean2 parameter.
comparator	string	or	Sets the comparator parameter.

math.convert_range

Prototype: core.node

Output: custom

Convert a value from one range to another.

Port	Type	Default	Description
value	float	50.0	Sets the value parameter.
source_start	float	0.0	Sets the source_start parameter.
source_end	float	100.0	Sets the source_end parameter.
target_start	float	0.0	Sets the target_start parameter.
target_end	float	1.0	Sets the target_end parameter.
method	string	clamp	Sets the method parameter.

math.coordinates

Prototype: core.node

Output: point

Calculate a new point based on the angle and distance from an original point.

Port	Type	Default	Description
position	point	0.00,0.00	Sets the position parameter.
angle	float	0.0	Sets the angle parameter.
distance	float	100.0	Sets the distance parameter.

math.cos

Prototype: core.node

Output: custom

Calculate the trigonometric cosine of an angle.

Port	Type	Default	Description
value	float	0.0	Sets the value parameter.

math.degrees

Prototype: core.node

Output: custom

Convert radians to degrees.

Port	Type	Default	Description
radians	float	0.0	Sets the radians parameter.

math.distance

Prototype: core.node

Output: custom

Calculate the distance between two points.

Port	Type	Default	Description
point1	point	0.00,0.00	Sets the point1 parameter.
point2	point	100.00,100.00	Sets the point2 parameter.

math.divide

Prototype: core.node

Output: custom

Divide two numbers.

Port	Type	Default	Description
value1	float	0.0	Sets the value1 parameter.
value2	float	1.0	Sets the value2 parameter.

math.e

Prototype: core.node

Output: custom

The value of the mathematical constant e, the base of the natural logarithm.

No input ports required (standalone generator).

math.even

Prototype: core.node

Output: boolean

Determine if a number is even.

Port	Type	Default	Description
value	int	0	Sets the value parameter.

math.floor

Prototype: core.node

Output: custom

Round down the value of a number.

Port	Type	Default	Description
value	float	0.0	Sets the value parameter.

math.integer

Prototype: core.node

Output: int

Create an integer value that can be used as a variable.

Port	Type	Default	Description
value	int	0	Sets the value parameter.

math.log

Prototype: core.node

Output: custom

Calculate the natural logarithm of a value.

Port	Type	Default	Description
value	float	1.0	Sets the value parameter.

math.make_numbers

Prototype: core.node

Output: custom

Transform a string to a list of numbers.

Port	Type	Default	Description
string	string	11;22;33	Sets the string parameter.
separator	string	;	Sets the separator parameter.

math.max

Prototype: core.node

Output: custom

Select the largest value from a list of numbers.

Port	Type	Default	Description
values	float	0.0	Sets the values parameter.

math.min

Prototype: core.node

Output: custom

Select the smallest value from a list of numbers.

Port	Type	Default	Description
values	float	0.0	Sets the values parameter.

math.mod

Prototype: core.node

Output: custom

Calculate the modulo by dividing two numbers and keeping the remainder.

Port	Type	Default	Description
value1	float	0.0	Sets the value1 parameter.
value2	float	1.0	Sets the value2 parameter.

math.multiply

Prototype: core.node

Output: custom

Multiply two numbers.

Port	Type	Default	Description
value1	float	0.0	Sets the value1 parameter.
value2	float	1.0	Sets the value2 parameter.

math.negate

Prototype: core.node

Output: custom

Change the sign of a number by negating it.

Port	Type	Default	Description
value	float	0.0	Sets the value parameter.

math.number

Prototype: core.node

Output: custom

Create a floating-point value that can be used as a variable.

Port	Type	Default	Description
value	float	0.0	Sets the value parameter.

math.odd

Prototype: core.node

Output: boolean

Determine if a number is odd.

Port	Type	Default	Description
value	int	0	Sets the value parameter.

math.pi

Prototype: core.node

Output: custom

The value of the mathematical constant pi.

No input ports required (standalone generator).

math.pow

Prototype: core.node

Output: custom

Calculate the power of a number.

Port	Type	Default	Description
value1	float	0.0	Sets the value1 parameter.
value2	float	0.0	Sets the value2 parameter.

math.radians

Prototype: core.node

Output: custom

Convert degrees to radians.

Port	Type	Default	Description
degrees	float	0.0	Sets the degrees parameter.

math.random_numbers

Prototype: core.node

Output: custom

Create a list of random numbers.

Port	Type	Default	Description
amount	int	10	Sets the amount parameter.
start	float	0.0	Sets the start parameter.
end	float	100.0	Sets the end parameter.
seed	int	0	Sets the seed parameter.

math.range

Prototype: core.node

Output: custom

Generate a range of numbers.

Port	Type	Default	Description
start	float	0.0	Starting number.
end	float	10.0	Ending limit.
step	float	1.0	Increment step size.

math.reflect

Prototype: core.node

Output: point

Calculate a new point based on the angle and distance from an original point.

Port	Type	Default	Description
point1	point	0.00,0.00	Sets the point1 parameter.
point2	point	100.00,100.00	Sets the point2 parameter.
angle	float	0.0	Sets the angle parameter.
distance	float	1.0	Sets the distance parameter.

math.round

Prototype: core.node

Output: int

Convert a floating-point number to an integer.

Port	Type	Default	Description
value	float	0.0	Sets the value parameter.

math.running_total

Prototype: core.node

Output: custom

Generate in between totals of a list of numbers.

Port	Type	Default	Description
values	float	0.0	Sets the values parameter.

math.sample

Prototype: core.node

Output: custom

Generate numbers within the given bounds.

Port	Type	Default	Description
amount	int	10	The number of samples to generate.
start	float	0.0	The start value of the sample sequence.
end	float	100.0	The end value of the sample sequence.

math.sin

Prototype: core.node

Output: custom

Calculate the trigonometric sine of an angle.

Port	Type	Default	Description
value	float	0.0	Sets the value parameter.

math.sqrt

Prototype: core.node

Output: custom

Calculate the square root of a number.

Port	Type	Default	Description
value	float	0.0	Sets the value parameter.

math.subtract

Prototype: core.node

Output: custom

Subtract two numbers.

Port	Type	Default	Description
value1	float	0.0	Sets the value1 parameter.
value2	float	0.0	Sets the value2 parameter.

math.sum

Prototype: core.node

Output: custom

Add up all numbers in the list.

Port	Type	Default	Description
values	float	0.0	Sets the values parameter.

math.wave

Prototype: core.node

Output: custom

Calculate a value based on wave equations.

Port	Type	Default	Description
min	float	0.0	Sets the min parameter.
max	float	100.0	Sets the max parameter.
period	float	60.0	Sets the period parameter.
offset	float	0.0	Sets the offset parameter.
type	string	sine	Sets the type parameter.

math.expression

Prototype: core.node

Output: custom

Evaluate a mathematical expression using variables x, y, z, t.

Port	Type	Default	Description
expression	string	$\sin(x) * 50$	Sets the expression parameter.
x	float	0.0	Sets the x parameter.
y	float	0.0	Sets the y parameter.
z	float	0.0	Sets the z parameter.
t	float	0.0	Sets the t parameter.

math.noise

Prototype: core.node

Output: custom

Generate 2D/3D Simplex, Worley, or Fractal Brownian Motion noise.

Port	Type	Default	Description
x	float	0.0	X coordinate in noise space.
y	float	0.0	Y coordinate in noise space.
z	float	0.0	Z coordinate / time evolution in noise space.
type	string	simplex	Sets the type parameter.
octaves	int	1	Fractal Brownian motion (fBm) octaves.
roughness	float	0.5	Sets the roughness parameter.

math.spring

Prototype: core.node

Output: custom

Simulate a damped spring oscillator.

Port	Type	Default	Description
current	float	0.0	Sets the current parameter.
target	float	100.0	Sets the target parameter.
velocity	float	0.0	Sets the velocity parameter.
tension	float	150.0	Sets the tension parameter.
damping	float	10.0	Friction damping factor.
mass	float	1.0	Sets the mass parameter.

NETWORK Library

4 Nodes

network.encode_url

Prototype: core.node

Output: custom

Convert text to a representation suitable for a URL.

Port	Type	Default	Description
value	string	Alpha & Omega	The text string to be converted.

network.http_get

Prototype: core.node

Output: custom

Get the contents of an URL.

Port	Type	Default	Description
url	string	https://nodebox.net/	The URL to fetch. Should start with 'http://' or 'https://.'
username	string	—	Username for authenticated requests (optional).
password	string	—	Password for authenticated requests (optional).
refreshTime	int	300	Amount of seconds before the resource is fetched again (time-to-live).

network.query_json

Prototype: core.node

Output: custom

Query the JSON string using JSONPath.

Port	Type	Default	Description
json	string	<pre>{"name": {"firstName": "Lois", "lastName": "Lane"}}}</pre>	The JSON data, as a string.
query	string	<code>\$.name.firstName</code>	The JSON path query.

network.fetch_json

Prototype: core.node

Output: custom

Fetch a JSON resource from a URL and extract data using JSONPath.

Port	Type	Default	Description
url	string	<code>https://api.github.com/zen</code>	The URL to fetch JSON from.
query	string	<code>\$</code>	The JSON path query (default \$ for root).
refreshTime	int	<code>300</code>	Cache time-to-live in seconds.

STRING Library

20 Nodes

string.as_binary_list

Prototype: core.node

Output: string

Convert a string into its binary representation

Port	Type	Default	Description
string	string	default	Sets the string parameter.

string.as_binary_string

Prototype: core.node

Output: string

Convert a string into its binary representation

Port	Type	Default	Description
string	string	default	Sets the string parameter.
digit_separator	string	—	Sets the digit_separator parameter.
byte_separator	string		Sets the byte_separator parameter.

string.as_number_list

Prototype: core.node

Output: string

Convert a string into a numeric encoding

Port	Type	Default	Description
string	string	default	Sets the string parameter.
radix	int	10	Sets the radix parameter.
padding	boolean	true	Sets the padding parameter.

string.change_case

Prototype: core.node

Output: string

Change a string into upper, lower or title case.

Port	Type	Default	Description
string	string	default	Sets the string parameter.
method	string	uppercase	Sets the method parameter.

string.character_at

Prototype: core.node

Output: string

Output the character at a given index

Port	Type	Default	Description
string	string	default	Sets the string parameter.
index	int	0	Sets the index parameter.

string.characters

Prototype: core.node

Output: string

Create a list of characters from a string

Port	Type	Default	Description
string	string	default	Sets the string parameter.

string.concatenate

Prototype: core.node

Output: string

Add strings together.

Port	Type	Default	Description
string1	string	–	Sets the string1 parameter.
string2	string	–	Sets the string2 parameter.
string3	string	–	Sets the string3 parameter.
string4	string	–	Sets the string4 parameter.
string5	string	–	Sets the string5 parameter.
string6	string	–	Sets the string6 parameter.
string7	string	–	Sets the string7 parameter.

string.contains

Prototype: core.node

Output: boolean

Determine if the string contains a given string

Port	Type	Default	Description
string	string	default	Sets the string parameter.
contains	string	efa	Sets the contains parameter.

string.ends_with

Prototype: core.node

Output: boolean

Determine if the string ends with a given string

Port	Type	Default	Description
string	string	default	Sets the string parameter.
ends_with	string	1t	Sets the ends_with parameter.

string.equals

Prototype: core.node

Output: boolean

Determine if the string equals a given string

Port	Type	Default	Description
string	string	default	Sets the string parameter.
equals	string	default	Sets the equals parameter.
case_sensitive	boolean	false	Sets the case_sensitive parameter.

string.format_number

Prototype: core.node

Output: string

Represent a numeric value as a string in a specific way.

Port	Type	Default	Description
value	float	0.0	Sets the value parameter.
format	string	%.2f	Sets the format parameter.

string.length

Prototype: core.node

Output: int

Count the number of characters in a string.

Port	Type	Default	Description
string	string	–	Sets the string parameter.

string.make_strings

Prototype: core.node

Output: string

Create a list of strings.

Port	Type	Default	Description
string	string	Alpha;Beta;Gamma	Sets the string parameter.
separator	string	;	Sets the separator parameter.

string.random_character

Prototype: core.node

Output: string

Create a list of random characters taken from a set

Port	Type	Default	Description
characters	string	abcdefghijklmnopqrstuvwxyzABCDEFGHIJKLMNOPQRSTUVWXYZ0123456789	Sets the characters parameter.
amount	int	10	Sets the amount parameter.
seed	int	0	Sets the seed parameter.

string.replace

Prototype: core.node

Output: string

Replace part of a string

Port	Type	Default	Description
string	string	default	Sets the string parameter.
old	string	AUL	Sets the old parameter.
new	string	aul	Sets the new parameter.

string.starts_with

Prototype: core.node

Output: boolean

Determine if the string starts with a given string

Port	Type	Default	Description
string	string	default	Sets the string parameter.
starts_with	string	def	Sets the starts_with parameter.

string.string

Prototype: core.node

Output: string

Create a string value that can be used as a variable.

Port	Type	Default	Description
value	string	–	Sets the value parameter.

string.sub_string

Prototype: core.node

Output: string

Take a portion of a string

Port	Type	Default	Description
string	string	default	Sets the string parameter.
start	int	0	Sets the start parameter.
end	int	4	Sets the end parameter.
end_offset	boolean	false	Sets the end_offset parameter.

string.trim

Prototype: core.node

Output: string

Remove white space from the start and end

Port	Type	Default	Description
string	string	def au lt	Sets the string parameter.

string.word_count

Prototype: core.node

Output: int

Count the number of words in a string.

Port	Type	Default	Description
string	string	–	Sets the string parameter.

4. Practical Creative Scenarios & Network Blueprints

Combine nodes into production-ready procedural workflows using these recipes:

Recipe 1: Moroccan Rosette & Seamless Wallpaper Pattern

corevector.ellipse → corevector.kaleidoscope → corevector.seamless_tile → corevector.colorize

Goal: Generate intricate, symmetrical tileable surface patterns for textiles and wallpaper.

1. Create a base petal with `corevector.ellipse` (width: 80, height: 28, position: 40, 0).
2. Connect into `corevector.kaleidoscope`. Set segments to 12 and mirror to true. This creates a 12-fold rosette with alternating bilateral reflections.
3. Pipe into `corevector.seamless_tile` (width: 300, height: 300, clip: false). The node detects all rosette paths that overlap the tile boundary and generates matching wraparound copies on opposite borders.
4. Colorize with `corevector.colorize` using a vibrant gradient from `color.gradient`.
5. Press **T** in the viewport to enable 3×3 infinite repeat and verify seamless alignment across all tile edges!

Recipe 2: Golden Ratio Phyllotaxis Procedural Duplicator

`corevector.rect` → `corevector.duplicator` → `color.gradient` → `corevector.colorize`

Goal: Generate natural sunflower seeds or pinecone spiral distributions with progressive step transformations.

1. Create a seed element using `corevector.rect` (width: 30, height: 10).
2. Connect into `corevector.duplicator`. Select mode = *spiral*.
3. Set count = 120, growthRate = 2.5, and angleStep = 137.5 (the Golden Angle).
4. Add step dynamics: set `stepRotation` = 6.0°, `stepScale` = 1.2%, and `stepOpacity` = 0.95.
5. Feed a multi-stop color gradient from `color.gradient` into the fill port of `corevector.colorize` to map color smoothly across the spiral depth.

Recipe 3: Audio-Reactive Multi-Band Frequency Spectrum Visualizer

`device.sound_file` → `device.sound_spectrum` → `corevector.rect` → `device.beat_detect` → `corevector.transform`

Goal: Create responsive motion graphics synchronized directly to an audio file (.wav, .mp3, .flac).

1. Load an audio file into `device.sound_spectrum` (set bands = 32). This outputs 32 normalized frequency magnitude values.
2. Connect the spectrum list to the height port of `corevector.rect`. Use `math.multiply` (factor: 300) so frequencies control bar heights.
3. Generate horizontal spacing with `corevector.grid` (rows: 1, columns: 32, width: 700).
4. Add `device.beat_detect` to listen to live kick drum beats; connect its pulse output to a `corevector.transform` scale port for bass-driven camera punch!

Recipe 4: Organic Perlin Noise Flow Field Typography Poster

`corevector.grid` → `corevector.flow_field` → `corevector.stroke_style` → `corevector.fit_text` → `corevector.long_shadow`

Goal: Produce editorial generative posters with fluid streamlines and bold 3D typography.

1. Generate a 40×40 seed point grid with `corevector.grid`.
2. Connect points to `corevector.flow_field` (steps: 60, stepLength: 4.0, noiseScale: 0.004).
3. Style the streamlines with `corevector.stroke_style` (dashArray: "8, 4", cap: round, join: round).
4. Overlay title typography with `corevector.fit_text` (width: 600, height: 36, alignment: CENTER).
5. Extrude dramatic cast shadows from the letters using `corevector.long_shadow` (distance: 120, angle: 45°).

Recipe 5: Geometric Branding with Variable Corner Rounding

`corevector.star` → `corevector.round_corners` → `corevector.extrude_3d` → `corevector.colorize`

Goal: Produce contemporary, smooth identity logos with chamfers, fillets, and squircles.

1. Create an 8-point star with `corevector.star` (outer: 150, inner: 60).
2. Connect into `corevector.round_corners`. Set `radius = 24` and `type = squircle` (G2 curvature continuous) or `chamfer` for crisp industrial facets.
3. Enable `clamp = true` to automatically prevent self-intersecting loops on tight interior vertices.
4. Give the logo isometric depth by feeding it into `corevector.extrude_3d` (depth: 25, angle: 30°, shading: true).

Recipe 6: Halftone Stippled Image Engraving

`data.sample_image` → `corevector.halftone` → `corevector.modular_grid` → `corevector.guilloche`

Goal: Convert raster images into printable, high-density vector dot halftones with banknote-style security guilloche borders.

1. Supply an image path to `corevector.halftone` (`gridSize: 8.0`, `maxDotSize: 7.5`).
2. The node samples pixel luminances and generates proportional vector circles.
3. Create ornamental banknote security guilloche rosettes with `corevector.guilloche` (`R: 120`, `r: 48`, `d: 35`, `steps: 800`) to frame the piece.

Recipe 7: Interactive Mouse Gravity Vortex

`device.mouse_position` → `corevector.attractor` → `corevector.wave_warp` → `corevector.colorize`

Goal: Build real-time interactive canvas graphics that warp and twist around the user's cursor.

1. Generate a concentric ring field with `corevector.ellipse` and `corevector.copy`.
2. Connect `device.mouse_position` to the `position` port of `corevector.attractor`.
3. Set `attractor.mode = vortex` and `force = 80`. As the mouse glides across the viewport, geometry spirals gravitationally into the cursor!
4. Deform vertices with `corevector.wave_warp` (`waveType: sine`, `axis: radial`) for ripple shockwaves.

Recipe 8: Multi-Artboard Cross-Platform Social Media Campaign

`corevector.artboard` → `list.combine` → **File** → **Export Artboards**

Goal: Render and batch-export identical generative artwork formatted across multiple aspect ratios.

1. Create the master artwork branch.
2. Feed the artwork into three separate `corevector.artboard` nodes:
 - Artboard 1: preset = *Instagram Story* (1080 × 1920).
 - Artboard 2: preset = *Instagram Post* (1080 × 1080).
 - Artboard 3: preset = *Twitter Header* (1500 × 500).
3. Set `clip = true` on each artboard so shapes are cleanly masked to each canvas format.
4. Combine all artboards with `list.combine` and use **File** → **Export Artboards** to render all marketing assets in one click!

5. Master Keyboard Shortcut & Hotkey Reference

Shortcut	Action	Context
G	Modal Grab / Move transform	Canvas Viewport
R	Modal Rotate transform	Canvas Viewport
S	Modal Scale transform	Canvas Viewport
X / Y	Constrain transform to X or Y axis	During Modal G, R, or S
Shift	Fine-precision transformation & snapping	During Modal G, R, S or Alt-drag
Alt + Left Drag	Direct parameter scrubbing on active node	Canvas Viewport
T	Toggle 3×3 Infinite Wallpaper Repeat preview	Canvas Viewport
Space + Drag	Pan viewport canvas	Canvas Viewport & Network View
Scroll Wheel	Smooth zoom centered at mouse cursor	Canvas Viewport & Network View
Ctrl / Cmd + 0	Reset view zoom and center to origin	Canvas Viewport & Network View
Ctrl / Cmd + Enter	Set active node as Render Root	Network View
F5 / Space	Play / Pause animation timeline	Global
Ctrl / Cmd + Z	Undo last action	Global
Ctrl / Cmd + Shift + Z	Redo last action	Global
Ctrl / Cmd + N	Quick-add node dialog	Network View
Delete / Backspace	Delete selected nodes or connections	Network View

Pro Tip: Hold **Alt** while dragging an output connection wire to duplicate and fan out connections to multiple destination ports simultaneously.